ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** biotic and abiotic components	All living and non-living parts interact.
2	**(c)** producers	Green plants fix energy.
3	**(a)** stratosphere	In the upper atmosphere.
4	**(a)** Temperature	Non-living factor.
5	**(a)** Grass, grasshopper, frog, snake	Producer to consumers in order.
6	**(a)** three oxygen atoms	O3.
7	**(c)** Green plants	They make food by photosynthesis.
8	**(c)** trophic levels	Each is a feeding level.
9	**(c)** harmful ultraviolet radiation	UV rays can cause skin cancer.
10	**(c)** consumers	Heterotrophs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Biotic: all living organisms (1); Abiotic: non-living factors like air, water, soil and temperature (1).	1+1
12	Each step of a food chain (1); Where the organisms obtain food in the same manner (1).	1+1
13	It absorbs harmful UV radiation of the Sun (1); It protects living beings (1).	1+1
14	Organisms that make their own food by photosynthesis (1); Green plants, algae (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Food chain: a single linear sequence of organisms (1); Food web: many interconnected food chains (1); Food web gives stability to the ecosystem (1).	1+1+1
16	They break down dead organisms (1); They return nutrients to the soil (1); They keep the environment clean (1).	1+1+1
17	Natural: forest, pond, lake (1); Artificial: crop fields, aquarium, garden (1); Natural has self-sustaining balance; artificial needs human care (1).	1+1+1
	OR: Sun is the source of energy (1); Producers trap solar energy in food (1); Energy passes along the chain, with a loss at each level (1).	1+1+1
18	Dead organisms and wastes would pile up (1); Nutrients would not be recycled (1); The producers would suffer for lack of nutrients (1).	1+1+1
	OR: Pesticides enter the food chain from crops and water (1); They are non-degradable and get stored in bodies (1); Concentration rises at higher trophic levels (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ecosystem: all living and non-living things interacting in an area (1); Biotic: producers, consumers, decomposers (1); Examples: plants, deer, fungi (1); Abiotic: air, water, soil, sunlight, temperature (1); Biotic components depend on abiotic components (1).	1+1+1+1+1
	OR: Increase in concentration of harmful chemicals along a food chain (1); Pesticide in water or crops (1); Consumed by herbivores and accumulates (1); Carnivores accumulate even more (1); Top consumers like humans have the highest level (1).	1+1+1+1+1
20	Producers make food by photosynthesis (1); Consumers: herbivores, carnivores, omnivores (1); Decomposers break down dead matter (1); Examples: grass, deer, lion (1); Bacteria and fungi are decomposers (1).	1+1+1+1+1
	OR: Biodegradable: broken down by microbes (1); Examples: food, paper, cotton (1); Non-biodegradable: plastic, glass, DDT (1); They persist for a long time (1); They pollute and harm living beings (1).	1+1+1+1+1

Mistakes to avoid:

- Drawing arrows in a food chain from the eater to the food; the arrow points from the organism that is eaten to the one that eats it.
- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.
- Counting decomposers as a trophic level of the food chain, or calling the Sun a producer; green plants are producers, the Sun is the source of energy.
- Confusing ozone (O_3) with oxygen (O_2), and writing that CO_2 (instead of CFCs) is the main cause of ozone depletion.
- Saying that all waste is non-biodegradable or that plastics decompose quickly; plastics, glass and DDT are non-biodegradable.
- Explaining biomagnification as increase of pesticide in producers only; the concentration is highest in the **top** consumers.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** producers	Plants and algae.
2	**(c)** chlorofluorocarbons (CFCs)	Used in refrigerators and aerosols.
3	**(a)** Fungi	They break down dead remains.
4	**(c)** second trophic level	Herbivores.
5	**(a)** Vegetable peels	Broken down by microbes.
6	**(b)** herbivores	Primary consumers.
7	**(b)** 10	The 10 per cent law.
8	**(c)** Plastic	Persists in the environment.
9	**(a)** Human beings	Eats both plants and animals.
10	**(a)** unidirectional	It flows from producers to consumers and is not reused.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Only 10% of the energy is passed to the next trophic level (1); The rest is lost as heat or used in life processes (1).	1+1
12	Accumulation of non-degradable chemicals in successive trophic levels (1); Concentration increases up the food chain (1).	1+1
13	Microorganisms like bacteria and fungi (1); Break down dead remains and wastes into simple substances, recycling nutrients (1).	1+1
14	Energy decreases at each successive level (1); Too little energy remains for higher levels (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Energy captured by producers passes to consumers (1); It cannot return to the Sun (1); Energy lost as heat is not reused (1).	1+1+1
16	Removal of one level disturbs the chain (1); Removing herbivores affects both plants and carnivores (1); Impact is greater for some levels than others (1).	1+1+1
17	Humans are at the top of the food chain (1); Pesticides get accumulated through biomagnification (1); Foods like grains, vegetables and milk carry them (1).	1+1+1
	OR: Used in respiration and movement (1); Lost as heat to the environment (1); Only about 10% is stored in the body (1).	1+1+1
18	Ozone is a molecule of three oxygen atoms (1); UV radiation splits O2 into atomic oxygen (1); Atomic oxygen combines with O2 to form O3 (1).	1+1+1
	OR: Non-biodegradable waste persists (1); Pollutes soil and water (1); Harms animals and creates health hazards (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Series of organisms linked by feeding (1); Grass, grasshopper, frog, snake (1); Producer, primary consumer, secondary consumer, tertiary consumer (1); Energy decreases up the levels (1); Only 10% passes on, so chains are short (1).	1+1+1+1+1
	OR: Waste can pollute air, soil and water (1); Plastic and chemicals harm animals (1); Solutions: reduce, reuse and recycle (1); Composting biodegradable waste (1); Segregation of waste at home (1).	1+1+1+1+1
20	Only 10% of energy is transferred to the next level (1); Rest lost as heat and used in life processes (1); Deer: 1000 J (1); Tiger: 100 J (1); Energy decreases at each trophic level (1).	1+1+1+1+1
	OR: Food web is a network of interconnected food chains (1); In a food chain, one path (1); In a food web, many alternate pathways (1); If one species is lost, others can take over (1); So the ecosystem remains stable (1).	1+1+1+1+1

Mistakes to avoid:

- Drawing arrows in a food chain from the eater to the food; the arrow points from the organism that is eaten to the one that eats it.
- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.
- Counting decomposers as a trophic level of the food chain, or calling the Sun a producer; green plants are producers, the Sun is the source of energy.
- Confusing ozone (O_3) with oxygen (O_2), and writing that CO_2 (instead of CFCs) is the main cause of ozone depletion.
- Saying that all waste is non-biodegradable or that plastics decompose quickly; plastics, glass and DDT are non-biodegradable.
- Explaining biomagnification as increase of pesticide in producers only; the concentration is highest in the **top** consumers.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** biodegradable	Natural decomposition.
2	**(d)** a crop field	Man-made, e.g. crop field or aquarium.
3	**(d)** a food web	Several chains.
4	**(a)** biomagnification	Accumulates in the food chain.
5	**(b)** a forest	Not man-made.
6	**(c)** 100 J	10 per cent of 10 per cent.
7	**(a)** Top carnivore	Biomagnification.
8	**(a)** the Sun	Producers capture solar energy.
9	**(c)** energy is lost at each step	Little energy remains at higher levels.
10	**(c)** waste management	Reducing pollution.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Biodegradable: broken down by microbes, e.g. paper (1); Non-biodegradable: not broken down, e.g. plastic (1).	1+1
12	A series of organisms in which food and energy pass from one to another (1); Grass, grasshopper, frog, snake (1).	1+1
13	Chlorofluorocarbons (1); Used as refrigerants and in aerosol sprays (1).	1+1
14	They are non-biodegradable (1); They cause pollution and harm animals (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Second level: 10% of 20,000 = 2000 J (1); Third level: 10% of 2000 (1); 200 J (1).	1+1+1
16	By CFCs (1); Ozone hole allows more UV to reach the earth (1); Causes skin cancer, cataract and damage to crops (1).	1+1+1
17	They are non-biodegradable (1); Block drains and cause water-logging (1); Harm animals when eaten (1).	1+1+1
	OR: They make food by photosynthesis (1); All consumers depend on them (1); They capture solar energy first (1).	1+1+1
18	Reduce the use of CFC-containing products (1); Use CFC-free refrigerants (1); Follow international agreements (UNEP) (1).	1+1+1
	OR: Reduce, reuse, recycle (1); Compost vegetable waste (1); Use cloth or jute bags (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sun is the source (1); Producers convert solar energy into chemical energy (1); Energy passes to consumers (1); It does not return to the previous level (1); Energy is lost as heat at each level (1).	1+1+1+1+1
	OR: O2 is split by UV into oxygen atoms (1); Atoms combine with O2 to form O3 (1); CFC releases chlorine atoms in the stratosphere (1); Chlorine breaks ozone into oxygen (1); Depletion lets harmful UV reach the earth (1).	1+1+1+1+1
20	UV radiation increases (1); Causes skin cancer and cataract (1); Chemicals: CFCs from refrigerators and aerosols (1); UNEP agreed in 1987 to freeze CFC production (1); Later CFC-free products were developed (1).	1+1+1+1+1
	OR: Plastics: non-degradable, so use alternatives and recycle (1); CFCs: use CFC-free products (1); Pesticides: biomagnification, so use organic farming (1); These have affected living beings (1); Awareness and rules are needed (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.
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